

Article

# Fabrication of Eco-Friendly Biomass-Based Hydrogel Beads as an Effective Matrix for Slow-Release Fertilizer

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## Abstract

Rapid dissolution of conventional fertilizers causes low nutrient-use efficiency and serious leaching losses, contributing to agricultural non-point source pollution. In this study, biomass-based slow-release fertilizer beads were prepared by ionic crosslinking of potato starch (ST), chitosan (CS), and corn-straw biochar (BC), using potassium nitrate ( $\text{KNO}_3$ ) as the model nutrient. The effects of ST/CS ratio and BC incorporation on bead structure, swelling, nutrient loading, release kinetics, and soil-column leaching were systematically investigated. Biochar incorporation formed a more compact and interconnected porous network and reduced the equilibrium swelling ratios of ST90/CS10, ST80/CS20, and ST70/CS30 from 188%, 176%, and 164% to 168%, 136%, and 104%, respectively. Although BC slightly decreased  $\text{KNO}_3$  loading capacity, it markedly slowed nutrient release; ST80/CS20/BC20 released 31.09%, 50.09%, and 81.82% of loaded  $\text{KNO}_3$  at 24, 72, and 504 h, respectively, which were 28.40%, 25.27%, and 11.30% lower than those of ST80/CS20. Kinetic fitting indicated that BC reduced the apparent release rate and promoted diffusion-controlled release behavior. Soil-column experiments further showed that the beads reduced  $\text{NO}_3^-$ -N and  $\text{K}^+$  leaching compared with free  $\text{KNO}_3$ , with ST80/CS20/BC20 showing the best balance between nutrient loading and release control. These results suggest that starch–chitosan–biochar beads are a promising biodegradable matrix for slow-release fertilizer applications.

**Keywords:** starch–chitosan hydrogel; biochar; potassium nitrate; slow-release fertilizer; nutrient leaching; sustainable agriculture



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## 1. Introduction

Efficient nutrient management is essential for maintaining crop productivity while reducing the environmental burden associated with intensive agriculture. Conventional soluble fertilizers, particularly nitrate- and potassium-based fertilizers, are rapidly dissolved after application and can migrate with irrigation or rainfall. According to the 2025 National Bulletin on Fertilizer Use Efficiency Monitoring released by the Ministry of Agriculture and Rural Affairs of China, fertilizer application to the three major grain crops in China (wheat, maize, and rice) was 3.1% higher in 2025 than in 2020, whereas the average use efficiencies of nitrogen, phosphorus, and potassium were only 43.3%, 28.9%, and 52.1%, respectively. Such low nutrient-use efficiency implies that a large proportion of applied nutrients may

be lost through leaching, runoff, volatilization, or fixation, causing economic losses and environmental risks such as soil acidification and water eutrophication [1].

Slow- and controlled-release fertilizers have therefore been developed to synchronize nutrient availability with crop demand. Common strategies include polymer coatings, inorganic coatings, matrix entrapment, and chemically regulated release systems [2,3]. However, some conventional coating materials, such as polyolefin-based polymers and urea–formaldehyde resins, may introduce concerns related to microplastic accumulation, low biodegradability, or residual by-products [4,5]. Recent advances have consequently shifted attention toward renewable and biodegradable carriers, including cellulose, sodium alginate, carrageenan, starch, chitosan, and their composites. Among these materials, starch is inexpensive, abundant, and biodegradable, whereas chitosan contains amino and hydroxyl groups that can participate in hydrogen bonding and ionic crosslinking, improving network integrity and nutrient retention.

Bio-based hydrogels are particularly attractive as fertilizer carriers because their three-dimensional hydrophilic networks can absorb water, provide diffusion resistance, and gradually release entrapped nutrients [6]. Previous studies have shown that polysaccharide hydrogels can improve soil water retention, reduce fertilizer dissolution rates, and promote nutrient uptake [7,8]. Nevertheless, many biomass-based hydrogels still require petrochemical modifiers, synthetic monomers, or additional reinforcing agents to achieve sufficient structural stability. Biochar offers a complementary solution because it contains stable carbon structures, mineral components, surface functional groups, porous domains, and cation-exchange sites [9]. In fertilizer systems, biochar has been used mainly as a coating material or as a nutrient adsorbent [10,11], while its integration into starch–chitosan hydrogel beads as an internal reinforcing and adsorption phase remains less explored. The potential advantage of such a composite system is that starch and chitosan provide biodegradable gel networks, whereas biochar increases pore tortuosity, strengthens the bead framework, and may further function as a residual soil amendment after polymer degradation.

In this study, potato starch (ST), chitosan (CS), and biochar (BC) were used as the primary raw materials, and sodium tripolyphosphate was employed as the crosslinking agent. Starch–chitosan–biochar composite hydrogel beads were fabricated via ionic crosslinking, using potassium nitrate as the model fertilizer, to develop a novel slow-release fertilizer system. Particular attention was given to the effect of biochar on the structure of the hydrogel beads and the slow-release behavior of potassium nitrate. The aim of this work was to develop a low-cost biomass-based carrier material using a relatively simple and environmentally friendly preparation process.

## 2. Materials and Methods

### 2.1. Materials

The biochar was derived from corn straw and prepared by pyrolysis at 750 °C. Its physicochemical properties are summarized in Table 1.

**Table 1.** Physicochemical properties of corn-straw-derived biochar.

| Feedstock  | Pyrolysis Conditions | C      | H     | N     | O     | BET Surface Area                         | Average Pore Diameter | Ash Content | pH   |
|------------|----------------------|--------|-------|-------|-------|------------------------------------------|-----------------------|-------------|------|
| Corn straw | 750 °C,<br>3 h       | 68.58% | 1.21% | 1.35% | 8.20% | 57.68<br>m <sup>2</sup> ·g <sup>−1</sup> | 1.406<br>nm           | 17.16%      | 9.38 |

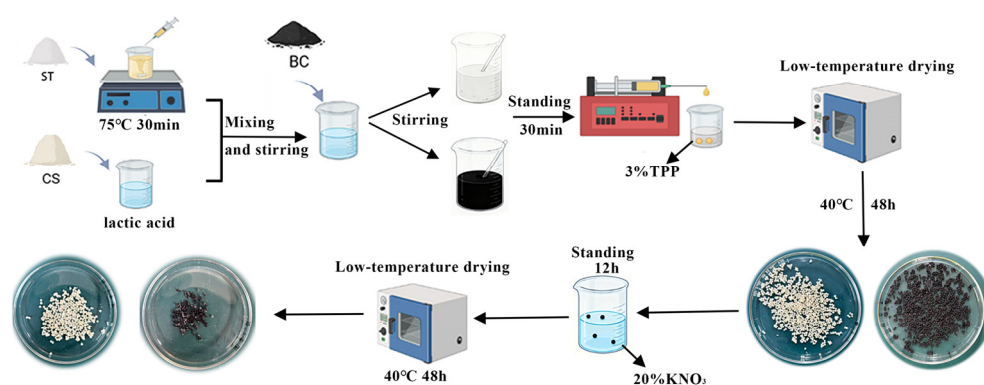
Note: BET, Brunauer-Emmett-Teller specific surface area.

Potato starch, chitosan (degree of deacetylation, 81%), sodium tripolyphosphate (purity, 91%), and potassium nitrate granules were purchased from Aladdin Biochemical

Technology Co., Ltd., Shanghai, China. All chemicals used in the experiments were of analytical grade.

## 2.2. Preparation of Biochar-Composite Hydrogel Beads

Biochar-composite hydrogel beads were prepared via ionic crosslinking (Figure 1). Potato starch was first dispersed in deionized water to prepare 9%, 8%, and 7% (*w/v*) starch suspensions, which were gelatinized at 76 °C for 30 min under magnetic stirring at 600 rpm to obtain gelatinized starch dispersions. Chitosan was dissolved in 1% (*v/v*) lactic acid aqueous solution to prepare 1%, 2%, and 3% (*w/v*) chitosan solutions. A 3% (*w/v*) sodium tripolyphosphate solution was prepared separately as the crosslinking agent. The ST/CS ratios of 90/10, 80/20, and 70/30 were selected to represent low, medium, and high chitosan contents while maintaining sufficient starch-based matrix formability. The BC loading was fixed at 20 wt% relative to the total ST + CS mass to allow a direct comparison between biochar-free and biochar-containing systems while avoiding an additional formulation variable. This level was selected to provide sufficient biochar-associated adsorption and diffusion regulation effects while maintaining acceptable dispersion and bead formability.



**Figure 1.** Schematic illustration of the preparation process of the hydrogel beads.

The starch and chitosan solutions were mixed on a dry weight basis to obtain ST/CS mass ratios of 90/10, 80/20, and 70/30. After mechanical stirring for 30 min, the mixtures were allowed to stand at room temperature for 30 min to remove entrapped air. Corn straw-derived biochar prepared by pyrolysis at 750 °C was then added at a fixed proportion of  $BC/(ST + CS) = 20$  wt%, resulting in ST/CS/BC mass ratios of 90/10/20, 80/20/20, and 70/30/20. The resulting suspensions were treated under the same stirring and standing conditions.

Each mixture was loaded into a 10 mL syringe and added dropwise into the sodium tripolyphosphate solution at a constant rate of  $3 \text{ mL min}^{-1}$  from a height of 10 cm above the liquid surface to form hydrogel beads. The beads were then crosslinked at room temperature for 4 h, collected, and washed three times with deionized water to remove residual crosslinker. Finally, the samples were dried at 40 °C for 48 h to obtain ST/CS/BC hydrogel beads. All treatments were conducted in triplicate unless otherwise stated.

## 2.3. Material Characterization

### 2.3.1. Scanning Electron Microscopy (SEM)

The surface morphology and microstructure of the samples were examined using a Phenom Pro X desktop scanning electron microscope (Thermo Fisher Scientific, Waltham, MA, USA). The biomass-composite hydrogel beads were mounted on aluminum stubs with double-sided adhesive tape and sputter-coated with gold before observation. Images were collected at magnifications ranging from  $150\times$  to  $5000\times$ .

### 2.3.2. Fourier Transform Infrared Spectroscopy (FTIR)

FTIR spectra of pure ST, CS, BC, and hydrogel beads were characterized using a Thermo Scientific Nicolet iS20 Fourier transform infrared spectrometer (Thermo Fisher Scientific, Waltham, MA, USA). The oven-dried hydrogel beads were ground into powder, thoroughly mixed with potassium bromide at a mass ratio of 1:100, and pressed into pellets for analysis.

### 2.3.3. X-Ray Diffraction (XRD)

The crystalline structure of the hydrogel beads was characterized using a Rigaku Smart Lab SE X-ray diffractometer (Rigaku Corporation, Tokyo, Japan). Measurements were conducted with Cu K $\alpha$  radiation at a tube voltage of 40 kV and a current of 20 mA. The incident X-ray wavelength was 0.154 nm, and diffraction patterns were recorded over a  $2\theta$  range of 5–60° at a scanning rate of 5° min<sup>−1</sup>.

### 2.3.4. Thermogravimetric Analysis (TGA)

The thermal stability of the hydrogel beads was evaluated using a TA Discovery SDT 650 simultaneous thermogravimetric analyzer (TA Instruments, New Castle, DE, USA). Approximately 10 mg of sample was heated from 50 to 800 °C at a rate of 10 °C min<sup>−1</sup> under a nitrogen atmosphere with a flow rate of 100 mL min<sup>−1</sup>.

## 2.4. Swelling Experiment

The hydrogel beads were dried to constant weight prior to the swelling test. Samples of known mass were then immersed in deionized water at room temperature. The equilibrium swelling ratio ( $Q$ ) was calculated according to the following equation:

$$Q = \frac{W_{t1}}{W_{t0}} \times 100\% \quad (1)$$

where  $W_{t1}$  denotes the weight (g) of the swollen beads at equilibrium, and  $W_{t0}$  represents the weight (g) of the dry polymer matrix before swelling.

## 2.5. Potassium Nitrate Loading

A 20 wt% potassium nitrate solution was prepared. The as-prepared ST/CS and ST/CS/BC hydrogel beads were separately immersed in the potassium nitrate solution and allowed to stand at room temperature for 12 h to achieve nutrient loading. After loading, the beads were collected and dried at 40 °C for 48 h to obtain low-temperature-dried starch-chitosan (ST/CS) and starch-chitosan-biochar (ST/CS/BC) fertilizer granules.

The nutrient loading capacity was calculated using the following equation:

$$\text{Loading capacity} = \frac{W_t - W_0}{W_0} \times 100\% \quad (2)$$

where  $W_t$  is the weight (g) of the hydrogel beads after loading, and  $W_0$  is the initial weight (g) of the hydrogel beads before loading.

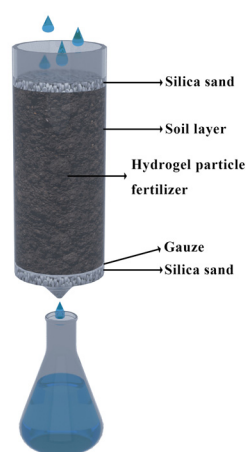
## 2.6. Static Water Release Experiment

The release behavior of KNO<sub>3</sub> from the loaded hydrogel beads was investigated using deionized water as the release medium. Six treatments were designed, namely ST90/CS10, ST80/CS20, ST70/CS30, ST90/CS10/BC20, ST80/CS20/BC20, and ST70/CS30/BC20, with three replicates for each treatment. Under static, non-stirring conditions, a fixed mass of loaded beads was placed in a steel basket and immersed in 100 mL of deionized water at 25 °C. At predetermined time intervals, the electrical conductivity of the release

medium was measured using a Hanna Instruments HI 9033 multi-range conductivity meter (Hanna Instruments, Woonsocket, RI, USA) to determine the cumulative concentration of released  $\text{KNO}_3$  [12], using corresponding beads without  $\text{KNO}_3$  loading as a control. To minimize interference from background ions released by the carrier matrix, the conductivity values were corrected using corresponding unloaded beads as blanks. Under these same background-corrected conditions, a calibration curve linking conductivity to  $\text{KNO}_3$  concentration was then established.

## 2.7. Soil Column Leaching Experiment

Ferralsol was used as the test soil. The soil was air-dried and passed through a 10-mesh sieve before use. A schematic diagram of the soil column apparatus is presented in Figure 2. The basic soil properties are summarized in Table 2. Seven treatments were included: free  $\text{KNO}_3$  granules, ST90/CS10, ST80/CS20, ST70/CS30, ST90/CS10/BC20, ST80/CS20/BC20, and ST70/CS30/BC20. Each treatment had three replicates, and the fertilizer dosage was standardized to  $1\text{g}\cdot\text{kg}^{-1}$  soil.



**Figure 2.** Schematic diagram of the soil column leaching experiment.

**Table 2.** Basic soil properties used in this study.

| pH<br>( $\text{H}_2\text{O}$ ) | EC<br>( $\mu\text{S}\cdot\text{cm}^{-1}$ ) | AK<br>( $\text{mg}\cdot\text{kg}^{-1}$ ) | AN<br>( $\text{mg}\cdot\text{kg}^{-1}$ ) | BD<br>( $\text{g}\cdot\text{cm}^{-3}$ ) | WHC<br>(%)       |
|--------------------------------|--------------------------------------------|------------------------------------------|------------------------------------------|-----------------------------------------|------------------|
| $4.68 \pm 0.01$                | $131.20 \pm 0.20$                          | $65.00 \pm 5.00$                         | $32.96 \pm 0.88$                         | $1.20 \pm 0.13$                         | $39.96 \pm 0.28$ |

Note: EC, electrical conductivity; AK, available potassium; AN, alkali-hydrolyzable nitrogen; BD, bulk density; WHC, water-holding capacity.

The leaching columns were made of white opaque polypropylene, with an inner diameter of 5 cm and a height of 25 cm. For column packing, two layers of 135-mesh gauze were first placed at the bottom, followed by 50 g of quartz sand as a support layer. The air-dried soil was thoroughly mixed with an equivalent amount of fertilizer granules and then packed in layers to a specified height at a bulk density of  $1.2\text{g}\cdot\text{cm}^{-3}$ . An additional 50 g of quartz sand was placed on top of each column to reduce soil disturbance during leaching.

After column preparation, deionized water was added to adjust the soil moisture content to 60% of the field water-holding capacity, and the columns were pre-incubated at  $25^\circ\text{C}$  in the dark. Leaching was conducted by adding 100 mL of water to each column on incubation days 1, 2, 4, 6, 8, 10, 15, and 26. The leachate generated within 24 h was collected, and the leachate volume was recorded. The concentration of nitrate nitrogen ( $\text{NO}_3^-$ -N) in the leachate was determined by ultraviolet spectrophotometry [13]. Orthophosphate ( $\text{PO}_4^{3-}$ -P) was determined using the molybdenum–antimony spectrophotometric method [14], and potassium ( $\text{K}^+$ ) was measured by flame atomic absorption spectrometry [15].

## 2.8. Release Kinetic Models

To gain further insight into the release mechanism of the hydrogel beads, the release data of the hydrogel beads and the corresponding biochar-composite systems were fitted using the Higuchi model, the Korsmeyer-Peppas model [16], and the first-order model [17]. The Higuchi model is described by the following equation:

$$\frac{M_t}{M_0} = k_H t^{\frac{1}{2}} \quad (3)$$

where  $M_t/M_0$  represents the cumulative release fraction at time  $t$ ,  $k_H$  is the release rate constant ( $\text{h}^{-1/2}$ ), and  $t$  denotes the release time.

The Korsmeyer-Peppas model is given by

$$\frac{M_t}{M_0} = k_{KP} t^n \quad (4)$$

where  $M_t/M_0$  represents the cumulative release fraction at time  $t$ ,  $k_{KP}$  is the release rate constant ( $\text{h}^{-1}$ ),  $n$  is the release exponent, and  $t$  denotes the release time. For the Korsmeyer-Peppas model, only the data points with  $M_t/M_0 \leq 0.6$  were used for fitting.

The first-order model is given by

$$\frac{M_t}{M_0} = 1 - e^{-k_1 t} \quad (5)$$

where  $M_t/M_0$  represents the cumulative release fraction at time  $t$ ,  $k_1$  is the release rate constant ( $\text{h}^{-1}$ ), and  $t$  denotes the release time.

## 2.9. Data Processing

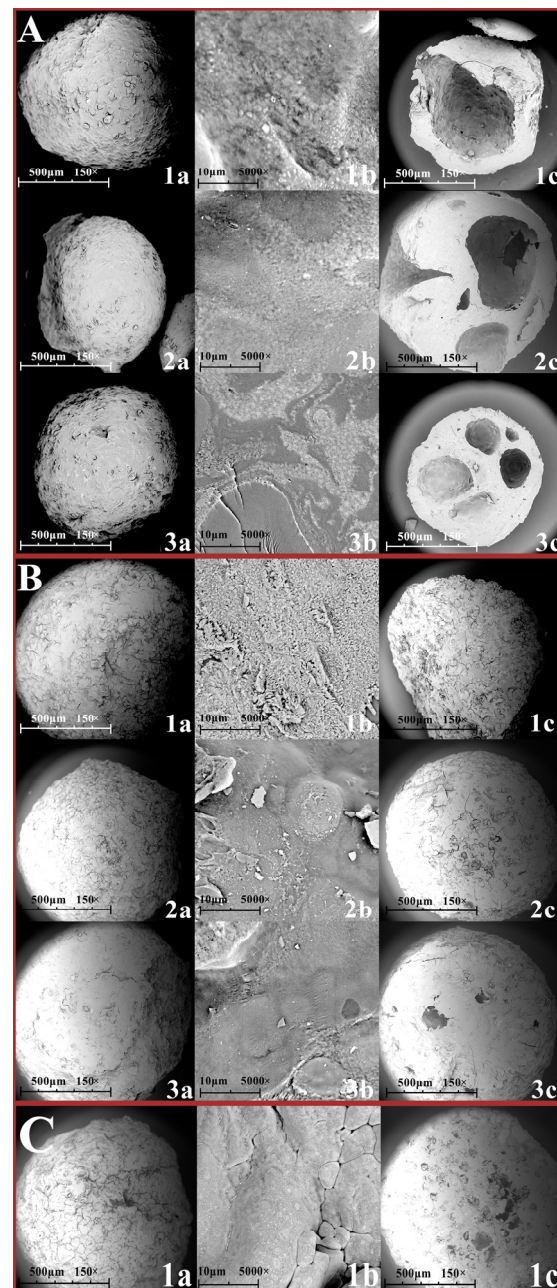
The values reported are the means and standard deviations of experiments ( $n = 3$ ). Data were analyzed by one-way ANOVA followed by Tukey's multiple comparison test, with  $p < 0.05$  considered statistically significant.

# 3. Results and Discussion

## 3.1. SEM Analyses

In this study, hydrogel beads with different starch-chitosan ratios and with or without biochar incorporation were prepared by ionic crosslinking. Their micromorphology and structural characteristics were examined by SEM to interpret the differences in macroscopic release behavior from a morphological perspective.

As shown in Figure 3A, the ST/CS ratio significantly influenced bead morphology and internal pore structure. The ST90/CS10 beads were spherical, but their cross-sections revealed large, poorly interconnected pores and local collapse (Figure 3A(1c)), indicating weak 3D network support. As the chitosan proportion increased, the bead surfaces became rougher, and the pore features more distinct (Figure 3A(3a,3b)), while pore-wall continuity improved and structural collapse defects were reduced in the cross-sections (Figure 3A(3c)). Both ST80/CS20 and ST70/CS30 beads displayed moderate surface wrinkling and roughness, as well as a well-developed, interconnected three-dimensional porous network in the cross-section. Such a structure can provide stable channels for the penetration of the release medium and solute transport. It also preserves a certain diffusion resistance, which may be structurally beneficial for a more sustained release profile [18,19].

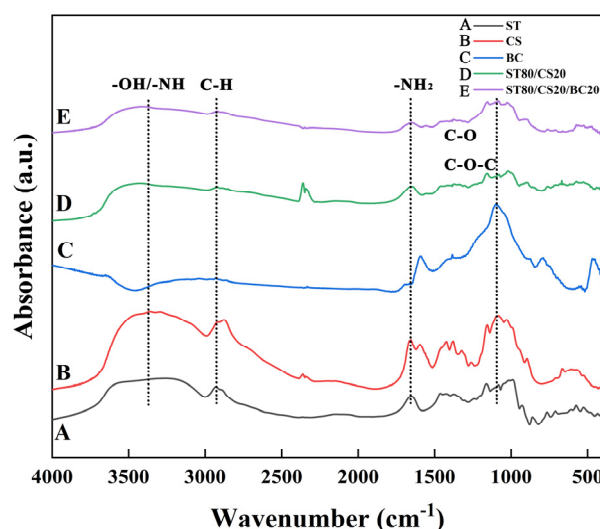


**Figure 3.** SEM images of hydrogel beads with different formulations: (A) ST/CS beads without biochar; (1a–1c) ST90/CS10; (2a–2c) ST80/CS20; (3a–3c) ST70/CS30; (B) ST/CS/BC composite beads containing corn straw-derived biochar; (1a–1c) ST90/CS10/BC20; (2a–2c) ST80/CS20/BC20; (3a–3c) ST70/CS30/BC20; (C) KNO<sub>3</sub>-loaded ST/CS/BC beads; (1a–1c) ST80/CS20/BC20 (a) Overall morphology at low magnification (150×); (b) surface morphology at high magnification (5000×); (c) cross-sectional structure (150×).

After biochar addition, the composite beads exhibited more pronounced debris-like or lamellar surface undulations (Figure 3B(1a,2a,3a,1b,2b,3b)). Their cross-sectional structures became more continuous and compact overall while retaining pore channels (Figure 3B(1c,2c,3c)), indicating that biochar reinforced the gel framework and modulated pore architecture [20,21]. In particular, ST80/CS20/BC20 showed biochar particles embedded within the matrix and a hierarchical pore network with pore sizes of approximately 100 nm to 10 µm (Figure 3B(2b,2c)). This morphology suggests a combined release-control mechanism: the polymer network provides a physical diffusion barrier, whereas biochar contributes adsorption sites and increases pathway tortuosity [22].

### 3.2. FTIR Analysis

Figure 4 shows the FTIR spectra of hydrogel beads with different formulations. Both potato starch and chitosan exhibited broad absorption bands in the range of  $3200\text{--}3600\text{ cm}^{-1}$ . For starch, the broad band at  $3200\text{--}3600\text{ cm}^{-1}$  was mainly assigned to O-H stretching. For chitosan, this region was associated with both O-H and N-H stretching vibrations. These bands indicate the presence of abundant hydrophilic functional groups in both materials. Potato starch showed a distinct peak at approximately  $1091\text{ cm}^{-1}$ . This peak corresponds to C-O and C-O-C vibrations, reflecting the typical polysaccharide backbone.



**Figure 4.** FTIR spectra of ST, CS, BC, ST80/CS20, and ST80/CS20/BC20.

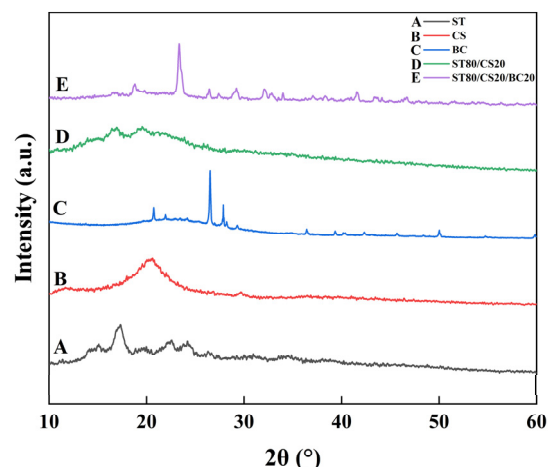
Chitosan showed a characteristic absorption near  $1659\text{ cm}^{-1}$ , which was associated with the amide I band and adsorbed water [23]. Biochar exhibited a relatively weak broadband in the high-wavenumber region, while a more evident absorption peak appeared at approximately  $1588.9\text{ cm}^{-1}$ , indicating the presence of aromatic carbon structures (C=C). Together with the overall spectral profile, this suggests that some oxygen-containing functional groups were still retained on the biochar surface [24,25].

The ST80/CS20 hydrogel beads retained the main characteristic absorption bands of potato starch and chitosan, although some peaks became broader and partially overlapped. In the ST/CS hydrogel beads, the broad band at  $3200\text{--}3600\text{ cm}^{-1}$  became less intense and broader. This change suggests hydrogen-bonding interactions between starch and chitosan. It also indicates the formation of an interconnected polymer network after compounding [26]. After biochar incorporation, ST80/CS20/BC20 maintained the basic characteristic absorptions of the ST/CS beads, while some peaks in the  $1600\text{--}1000\text{ cm}^{-1}$  region showed further changes. The absorption near  $1588.9\text{ cm}^{-1}$  was assigned to the C=C vibration of aromatic structures in biochar. This peak confirmed the successful incorporation of biochar into the composite beads. It also suggests possible interfacial interactions between biochar and the starch–chitosan matrix [27,28]. No obvious new characteristic peaks were observed. This indicates that biochar was mainly introduced through physical incorporation and intermolecular interactions, rather than through the formation of new covalent bonds.

Overall, the FTIR results suggest that the formation of ST/CS hydrogel beads resulted from the combined effects of TPP-mediated ionic crosslinking and intermolecular hydrogen bonding between starch and chitosan. The incorporation of biochar altered the local chemical environment of some functional groups but did not change the fundamental polysaccharide-based chemical framework of the composite system.

### 3.3. XRD Analysis

To clarify the effect of biochar incorporation on the crystalline structure of starch-chitosan hydrogel beads, X-ray diffraction (XRD) analysis was performed, and the results are shown in Figure 5. XRD is useful for distinguishing the crystalline and amorphous features of the system, thereby providing a crystallographic basis for interpreting structural differences in the composites and the associated changes in their macroscopic properties.



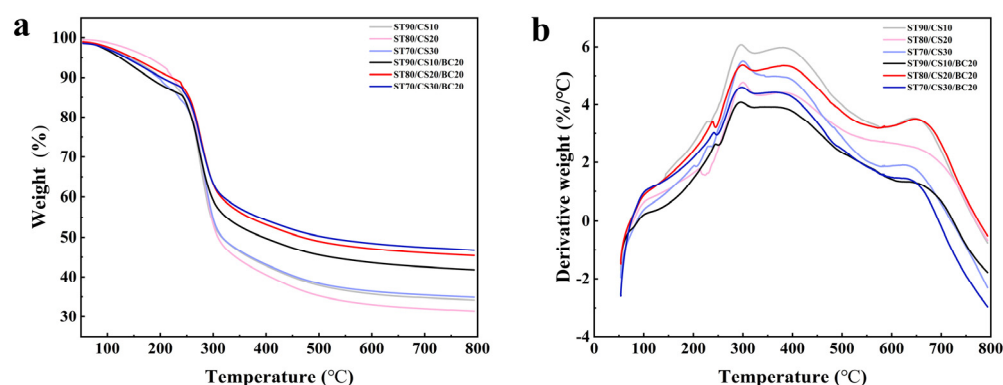
**Figure 5.** XRD patterns of ST, CS, BC, ST80/CS20, and ST80/CS20/BC20.

As shown in Figure 5, potato starch exhibited relatively clear diffraction peaks in the range of 15–25°, indicating a certain semi-crystalline character. Chitosan mainly presented a broad peak around 20°, suggesting low crystallinity and a predominantly amorphous structure. By contrast, biochar displayed several relatively sharp characteristic peaks, especially a pronounced peak near 26.52°, indicating the presence of relatively stable crystalline phases, which may originate from mineral components or locally ordered carbon structures.

Compared with the raw materials, the diffraction peaks of the ST80/CS20 hydrogel beads became markedly broader and weaker, and the pattern in the 2θ range of 15–23° was dominated by a broad diffuse peak. This suggests that the original crystalline regions of starch and chitosan were disrupted during gelatinization, blending, and gel formation, leading to decreased crystallinity and a more amorphous structure of the system [29]. After biochar incorporation, ST80/CS20/BC20 maintained the amorphous background while exhibiting several characteristic peaks attributable to biochar, confirming its successful incorporation into the composite bead system and the retention of part of its crystalline features. The pronounced diffuse background still observed in the composite sample indicates that the amorphous characteristics of the polymer hydrogel network remained dominant [30]. Overall, the gelled ST/CS system was dominated by an amorphous network, whereas biochar mainly acted as a functional filler embedded within the matrix without changing the fundamentally amorphous nature of the composite matrix.

### 3.4. TG Analysis

The thermogravimetric (TG) and derivative thermogravimetric (DTG) curves of the different samples are shown in Figure 6. All samples exhibited a multi-stage weight-loss pattern, including a gradual weight loss at low temperatures, a main decomposition stage at intermediate temperatures, and a slow carbonization/residual stage at high temperatures. This indicates a typical multi-step thermal decomposition process, while the biochar-containing composites reflected the combined thermal behavior of the polysaccharide network and the thermally stable filler [31].



**Figure 6.** (a) TG curves of different types of ST/CS and ST/CS/BC beads; (b) DTG curves of different types of ST/CS and ST/CS/BC beads.

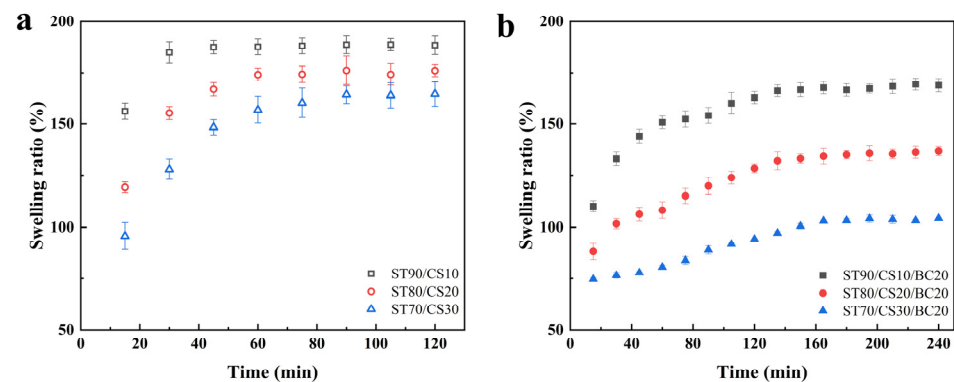
In the temperature range of 50–250 °C, the TG curves of all samples decreased gradually. Because the hydrogels contained abundant hydrophilic groups, the weight loss in this low-temperature region was mainly attributed to the removal of adsorbed water, bound water, and a small amount of volatile components [32]. Compared with the samples without biochar, the biochar-containing composites showed some differences in weight-loss behavior within this temperature range. These differences may be associated with the adsorption and retention of water by the porous structure of biochar.

In the range of 250–350 °C, the TG curves showed a pronounced decline, corresponding to the main weight-loss stage of the samples. Consistently, the DTG curves displayed a major decomposition peak around 300 °C (Figure 6b), indicating thermal degradation of organic components such as starch and chitosan, including cleavage of sugar rings and glycosidic bonds as well as decomposition of nitrogen-containing structures. After biochar incorporation, the samples retained the main decomposition features of the polymer network, but the weight-loss rate and residue formation in the medium- to high-temperature region changed. This suggests that biochar influenced the decomposition pathway and carbonization process through its thermally stable framework and interfacial interactions [33].

In the range of 350–800 °C, all samples continued to lose weight, although at a much lower rate, mainly corresponding to the further carbonization and structural rearrangement of residual organic components. Compared with the samples without biochar, the biochar-containing composites maintained a higher TG residue at high temperatures, indicating that the thermally stable carbon phase and ash components in biochar contributed to enhancing the high-temperature residue and reducing the overall mass loss of the system. The improved thermal stability also indirectly suggests that the composite system possessed greater structural stability [34]. Combined with the above morphological results, it can be inferred that such stability may help maintain longer diffusion pathways and more stable pore structures, thereby exerting a positive effect on the subsequent slow-release behavior.

### 3.5. Swelling and Nutrient Loading Analyses of Hydrogel Beads

Figure 7 illustrates the regulatory effects of ST/CS ratio and biochar incorporation on bead swelling. All samples exhibited typical swelling kinetics in deionized water, characterized by rapid initial water uptake followed by a gradual approach to equilibrium. This process reflects water diffusion into the bead matrix, relaxation of polymer chains, and the establishment of swelling equilibrium. The ST/CS ratio was a key factor controlling swelling capacity. As the chitosan proportion increased from 10% to 30%, the equilibrium swelling ratio decreased from approximately 188% to 164% (Table 3), indicating that higher chitosan content enhanced ionic crosslinking and network compactness, thereby restricting chain expansion and water uptake [35].



**Figure 7.** (a) Swelling ratios of ST/CS hydrogel beads with different formulations. (b) Swelling ratios of ST/CS/BC hydrogel beads with different formulations ( $n = 3$ ).

**Table 3.** Swelling ratios and potassium nitrate loading capacities of ST/CS and ST/CS/BC hydrogel beads with different formulations.

| Sample   | Drying Method | Ratio    | Swelling Ratio (%) | Loading Capacity (%) |
|----------|---------------|----------|--------------------|----------------------|
| ST/CS    | Oven drying   | 90/10    | 188                | 84.64                |
|          |               | 80/20    | 176                | 70.13                |
|          |               | 70/30    | 164                | 54.86                |
| ST/CS/BC | Oven drying   | 90/10/20 | 168                | 74.00                |
|          |               | 80/20/20 | 136                | 65.50                |
|          |               | 70/30/20 | 104                | 47.00                |

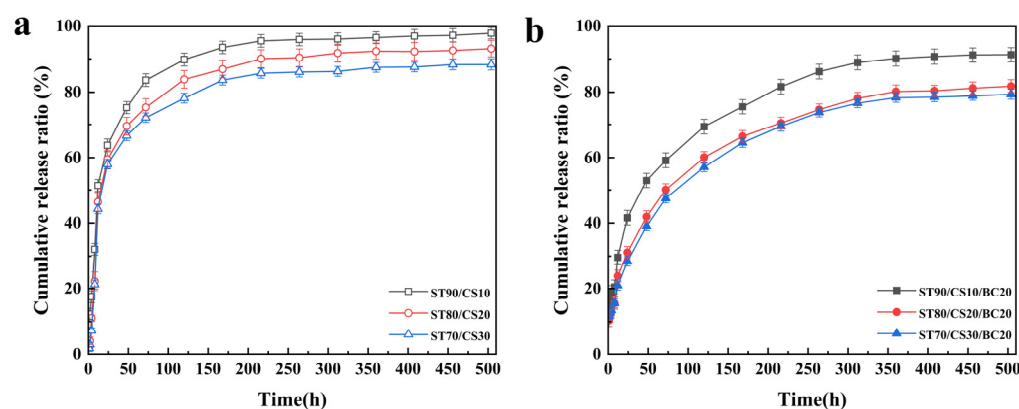
After biochar incorporation, the swelling curves shifted downward, and the time required to reach equilibrium was prolonged (Figure 7b). At the same ST/CS ratio, samples containing 20% biochar consistently showed lower equilibrium swelling ratios than their biochar-free counterparts. For example, ST90/CS10/BC20 decreased from 188% to 168%, ST80/CS20/BC20 from 176% to 136%, and ST70/CS30/BC20 from 164% to 104%. This reduction can be attributed to the rigid filling effect and interfacial constraint introduced by biochar, which reduced the effective free volume of the matrix and increased resistance to water transport. Thus, the ST/CS ratio regulated the primary network compactness, while biochar further reinforced the matrix and suppressed swelling [36].

The reduced swelling capacity was consistent with the lower KNO<sub>3</sub> loading capacity (Table 3). After biochar incorporation, the loading capacity decreased from 84.64% to 74.00% for ST90/CS10, from 70.13% to 65.50% for ST80/CS20, and from 54.86% to 47.00% for ST70/CS30. These results indicate that network densification and biochar-induced filling reduced accessible pore volume and solution penetration during impregnation. However, lower loading capacity should not be interpreted only as a disadvantage, because the denser and more tortuous structure simultaneously reduced nutrient diffusion and improved release control. The ST80/CS20/BC20 provided a favorable compromise that retained a relatively high loading capacity (65.50%) while showing stronger release retardation than ST80/CS20 and better loading than ST70/CS30/BC20.

### 3.6. Release Behavior in Static Water

Figure 8a shows that the static-water release behaviors of ST/CS beads followed a similar trend: relatively rapid release at the initial stage followed by a gradual approach to a plateau. However, the ST/CS ratio affected both the release rate and the final cumulative release. With increasing chitosan content, the cumulative release ratios for ST70/CS30 at 24, 72, and 504 h were 58.00%, 72.13%, and 90.19%, respectively, which were lower than

those of ST90/CS10 (63.73%, 83.65%, and 97.82%) and ST80/CS20 (59.49%, 75.35%, and 93.12%). These results indicate that ST70/CS30 formed a more compact network with stronger diffusion-retarding effects, while ST90/CS10 contained insufficient chitosan to provide comparable network constraint.



**Figure 8.** (a) Cumulative release profiles of ST/CS hydrogel beads with different mass ratios in static water. (b) Cumulative release profiles of ST/CS/BC hydrogel beads with different mass ratios in static water ( $n = 3$ ).

After BC incorporation, the release behavior of the beads was generally characterized by a markedly slower release during the middle and later stages. A lower final cumulative release was observed, compared with the corresponding samples without BC. As shown in Figure 8b, the BC-containing samples generally showed lower cumulative release ratios during the subsequent release process. For example, the cumulative release ratios of ST80/CS20/BC20 at 24, 72, and 504 h were 31.09%, 50.09%, and 81.82%, respectively, representing reductions of 28.40%, 25.27%, and 11.30% relative to those of ST80/CS20. Similarly, the cumulative release ratio of ST70/CS30/BC20 at 504 h was 79.38%, lower than that of ST70/CS30 (88.38%), while ST90/CS10/BC20 reached 91.30% at 504 h, also lower than that of ST90/CS10 (97.82%). These results indicate that the biochar addition significantly enhanced the release-retarding effect during the middle and later stages and effectively reduced the final cumulative release level. This behavior may be attributed to the porous structure and surface functional groups of biochar. They enhanced nutrient adsorption and retention while increasing the tortuosity of internal mass-transfer pathways, thereby suppressing the continued outward diffusion of nutrients [37]. Notably, the release profiles of ST80/CS20/BC20 and ST70/CS30/BC20 were similar, indicating that biochar incorporation retarded release more effectively than increasing chitosan content within the tested range.

### 3.7. Release Kinetics

The kinetic fitting results are summarized in Table 4. The release processes of the different samples did not strictly follow a single controlling mechanism. The incorporation of biochar clearly altered the characteristics of the kinetic parameters. For the ST/CS beads without biochar, the Higuchi model showed relatively low goodness-of-fit values, with  $R^2$  ranging from 0.53 to 0.65, indicating that the overall release process could not be simply described as being governed solely by Fickian diffusion. In contrast, the first-order model provided a markedly better fit, with  $R^2$  values ranging from 0.87 to 0.96. In particular, the first-order  $R^2$  value for ST90/CS10 reached 0.96, indicating that its overall release process was more strongly dependent on the concentration gradient. In terms of the  $k_1$  values, ST70/CS30 exhibited the lowest  $k_1$  value ( $0.025 \text{ h}^{-1}$ ) among the biochar-free systems, which was lower than those of ST80/CS20 ( $0.029 \text{ h}^{-1}$ ) and ST90/CS10 ( $0.042 \text{ h}^{-1}$ ). This

is consistent with its lower cumulative release ratios at 24, 72, and 504 h, indicating that the ST70/CS30 formulation had the lower apparent release rate and the best slow-release performance among the biochar-free systems.

**Table 4.** Parameters of the release kinetic models for different types of hydrogel beads.

| System/Formulation | Model       | Parameters                                   | $R^2$ |
|--------------------|-------------|----------------------------------------------|-------|
| ST90/CS10          | Higuchi     | $k_H = 0.056 \text{ h}^{-1/2}$               | 0.52  |
|                    | K-P         | $k_{KP} = 0.080 \text{ h}^{-1}$ . $n = 0.69$ | 0.95  |
|                    | First-order | $k_1 = 0.042 \text{ h}^{-1}$                 | 0.96  |
| ST80/CS20          | Higuchi     | $k_H = 0.051 \text{ h}^{-1/2}$               | 0.65  |
|                    | K-P         | $k_{KP} = 0.021 \text{ h}^{-1}$ . $n = 1.13$ | 0.85  |
|                    | First-order | $k_1 = 0.029 \text{ h}^{-1}$                 | 0.93  |
| ST70/CS30          | Higuchi     | $k_H = 0.050 \text{ h}^{-1/2}$               | 0.65  |
|                    | K-P         | $k_{KP} = 0.015 \text{ h}^{-1}$ . $n = 1.21$ | 0.86  |
|                    | First-order | $k_1 = 0.025 \text{ h}^{-1}$                 | 0.87  |
| ST90/CS10/BC20     | Higuchi     | $k_H = 0.049 \text{ h}^{-1/2}$               | 0.82  |
|                    | K-P         | $k_{KP} = 0.130 \text{ h}^{-1}$ . $n = 0.34$ | 0.96  |
|                    | First-order | $k_1 = 0.012 \text{ h}^{-1}$                 | 0.85  |
| ST80/CS20/BC20     | Higuchi     | $k_H = 0.042 \text{ h}^{-1/2}$               | 0.89  |
|                    | K-P         | $k_{KP} = 0.086 \text{ h}^{-1}$ . $n = 0.39$ | 0.97  |
|                    | First-order | $k_1 = 0.006 \text{ h}^{-1}$                 | 0.82  |
| ST70/CS30/BC20     | Higuchi     | $k_H = 0.043 \text{ h}^{-1/2}$               | 0.90  |
|                    | K-P         | $k_{KP} = 0.092 \text{ h}^{-1}$ . $n = 0.36$ | 0.97  |
|                    | First-order | $k_1 = 0.006 \text{ h}^{-1}$                 | 0.83  |

Note:  $k_H$ ,  $k_{KP}$ , and  $k_1$  are the rate constants of the Higuchi, Korsmeyer–Peppas (K-P), and first-order models, respectively;  $n$  is the release exponent in the K-P model.

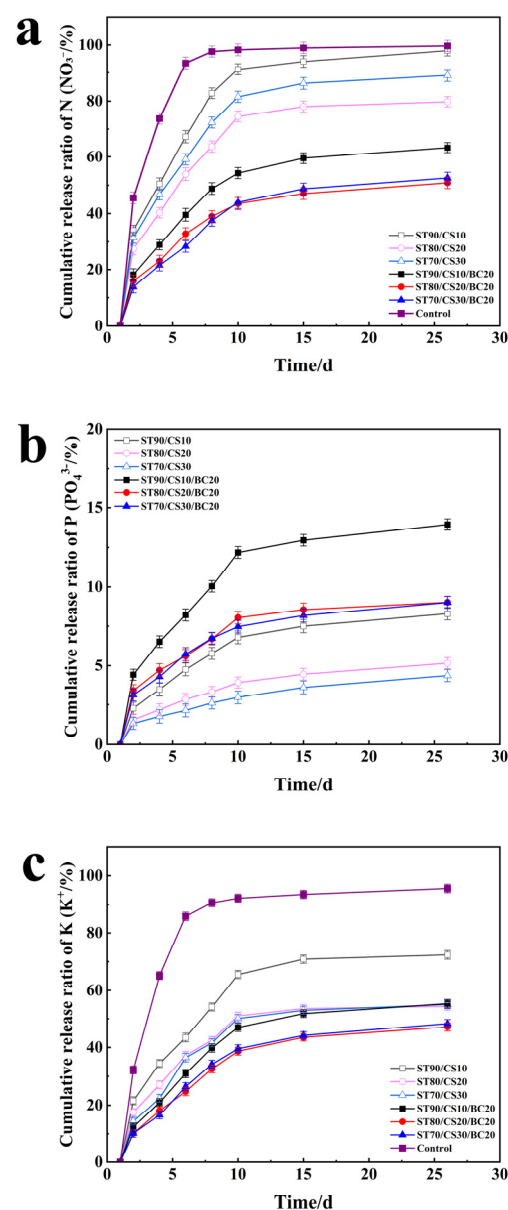
When the Korsmeyer–Peppas model was applied to the initial release stage ( $M_t/M_0 \leq 0.6$ ), the  $n$  values of ST90/CS10, ST80/CS20, and ST70/CS30 in the biochar-free system were 0.69, 1.13, and 1.21, respectively. ST90/CS10 was classified as exhibiting anomalous transport. By contrast, ST80/CS20 and ST70/CS30 showed  $n$  values greater than 0.85. This suggests that the early-stage release from these beads was not governed solely by Fickian diffusion. Rather, it involved multiple processes, including solute diffusion, water-induced swelling, polymer chain relaxation, and local structural rearrangement [38]. By comparison, after biochar incorporation, the  $R^2$  values of the Higuchi model increased to 0.82–0.90, which were higher than those of the corresponding biochar-free systems. At the same time, the  $k_H$  values decreased from 0.050–0.056  $\text{h}^{-1/2}$  in the biochar-free systems to 0.042–0.049  $\text{h}^{-1/2}$ , indicating that biochar incorporation reduced the apparent diffusion rate and caused the release process to exhibit more pronounced diffusion-limited characteristics.

More notably, the release exponent  $n$  of the biochar-containing samples decreased to 0.34–0.39, suggesting that the early-stage release mechanism shifted from the original combined mass-transfer behavior toward a more evidently diffusion-controlled process after biochar addition [39]. However, from the perspective of the entire release process, the first-order  $R^2$  values of the biochar-containing systems were 0.82–0.85. These values were lower than those of the corresponding biochar-free systems. This indicates that the release behavior became more complex after biochar incorporation and could no longer be adequately described by a single concentration-dependent decay model. Overall, among the biochar-free systems, ST70/CS30 exhibited the lowest first-order rate constant and the best overall slow-release performance. After biochar addition, ST70/CS30/BC20 and ST80/CS20/BC20 maintained lower final cumulative release and lower  $k_1$  values than ST90/CS10/BC20. Specifically, their cumulative release ratios at 504 h were 79.38% and 81.82%, respectively, both lower than that of ST90/CS10/BC20 (91.30%), while their  $k_1$

values were  $0.0057 \text{ h}^{-1}$  and  $0.0064 \text{ h}^{-1}$ , respectively, lower than that of ST90/CS10/BC20 ( $0.0120 \text{ h}^{-1}$ ). These results indicate that the combination of higher chitosan content and biochar incorporation contributed to stronger sustained-release regulation.

### 3.8. Release Behavior in Soil

The cumulative leaching rates of nitrogen ( $\text{NO}_3^-$ -N), phosphorus ( $\text{PO}_4^{3-}$ -P), and potassium ( $\text{K}^+$ ) were continuously monitored during the incubation period to evaluate the slow-release and nutrient-retention performance of hydrogel beads with different ST/CS ratios and biochar incorporation under simulated soil conditions. The results are presented in Figure 9, with pure  $\text{KNO}_3$  granules used as the control. Overall, the control treatment consistently showed the highest cumulative N and K leaching rates, indicating that unprotected soluble nutrients migrated rapidly after water input. In contrast, all hydrogel bead treatments reduced nutrient leaching to different extents, demonstrating that the bead network and interfacial characteristics retarded nutrient release and migration.



**Figure 9.** Leaching profiles of different types of ST/CS and ST/CS/BC hydrogel beads in soil columns: (a) N, (b) P, and (c) K ( $n = 3$ ).

As shown in Figure 9a, the nitrogen ( $\text{NO}_3^-$ -N) leaching ratio of the control approached 100% within 6–10 d, confirming the high mobility and leaching risk of nitrate. All bead treatments reduced cumulative N leaching. Among biochar-free treatments, ST90/CS10 showed the highest N leaching ratio (approximately 98%), whereas ST80/CS20 and ST70/CS30 showed lower values. After biochar addition, N leaching decreased further, with ST80/CS20/BC20 and ST70/CS30/BC20 showing the lowest values, both around 50%. These results suggest that higher chitosan content and biochar incorporation suppressed nitrate migration by increasing network compactness, creating additional interfacial retention sites, and extending diffusion pathways [40].

The leaching behavior of K ( $\text{K}^+$ ) was generally consistent with that of N, with the control showing the highest leaching rate and the composite bead treatments showing reduced leaching. The control rapidly reached a high cumulative leaching level within 6–10 days, whereas both ST/CS and ST/CS/BC beads delayed the release and migration of  $\text{K}^+$ . Among the biochar-free treatments, ST90/CS10 showed the highest  $\text{K}^+$  leaching rate, indicating that the weaker network constraint associated with lower chitosan content allowed  $\text{K}^+$  to migrate more readily with water flow. In addition, hydrophilic functional groups in the hydrogel, such as -OH and  $-\text{NH}_2$ , may provide a certain degree of  $\text{K}^+$  retention through weak electrostatic interactions or coordination/hydrogen-bond-related effects, thereby further slowing its diffusion [41]. After biochar incorporation, the cumulative leaching rates of  $\text{K}^+$  decreased overall, and the inhibitory effects were especially pronounced for ST80/CS20/BC20 and ST70/CS30/BC20. This suggests that, for the cationic nutrient  $\text{K}^+$ , biochar incorporation not only hinders mass transfer through pore-structure effects but may also enhance  $\text{K}^+$  retention through surface functional groups and ion-exchange sites [42], thereby producing a marked reduction in leaching loss.

Compared with N and K, the leaching rate of phosphorus ( $\text{PO}_4^{3-}$ -P) was much lower throughout the incubation period. Since  $\text{KNO}_3$  was used as the model fertilizer and does not contain phosphorus, the detected phosphorus in the leachate was most likely associated with residual or releasable sodium tripolyphosphate used during bead preparation [43]. Therefore, the phosphorus result should be interpreted as an auxiliary observation related to crosslinker-derived phosphorus migration rather than as a primary indicator of fertilizer-release performance.

Overall, the leaching rates of N and K generally decreased with increasing chitosan content and with biochar incorporation. Higher chitosan content likely increased ionic crosslinking density and reduced effective pore size, whereas biochar increased tortuosity and provided additional ion-retention sites. The trend for phosphorus was less direct because P originated from the TPP crosslinker and could be affected by residual crosslinker removal, bead degradation, and interactions with soil and biochar surfaces. These factors explain why P behavior should not be over-interpreted as fertilizer nutrient release in the present  $\text{KNO}_3$ -loaded system.

These soil-column results demonstrate that the composite hydrogel beads markedly reduced early N and K leaching and that biochar-containing treatments generally showed better leaching reduction than the corresponding biochar-free treatments. Combined with SEM, swelling, and static-water release results, the main factors responsible for nutrient-leaching suppression were matrix densification, adsorption/retention effects introduced by biochar, and elongation of mass-transfer pathways [44]. Among the tested formulations, ST80/CS20/BC20 and ST70/CS30/BC20 showed superior control of N and K leaching, while ST80/CS20/BC20 retained a higher loading capacity. Therefore, ST80/CS20/BC20 is recommended as the balanced formulation for further agronomic evaluation.

### 3.9. Comparison with Related Systems and Practical Implications

Compared with conventional coated fertilizers, the ST/CS/BC beads avoid persistent synthetic coatings by using renewable starch, chitosan, and biomass-derived biochar. Their novelty lies in embedding biochar within the starch–chitosan hydrogel matrix, rather than using it only as an external coating or nutrient adsorbent. This structure combines polymer-network diffusion resistance with biochar-related adsorption and ion-exchange effects, thereby improving nutrient retention and slowing release [45].

The soil-column results suggest that ST80/CS20/BC20 and ST70/CS30/BC20 can reduce  $\text{NO}_3^-$ -N and  $\text{K}^+$  leaching under simulated irrigation, indicating potential for reducing fertilizer-related nutrient losses. However, field validation, crop-yield assessment, mechanical-strength testing, and long-term degradation studies are still required before practical application. These factors should be systematically evaluated in future work to better clarify nutrient-retention mechanisms and practical applicability during storage, transport, and field application.

## 4. Conclusions

In this study, starch/chitosan/biochar hydrogel beads were successfully fabricated by TPP-mediated ionic crosslinking as biodegradable matrix-type carriers for slow-release  $\text{KNO}_3$  fertilizer. Biochar acted as a functional internal filler within the starch–chitosan network, contributing to improved structural stability and increased diffusion-pathway tortuosity. Although biochar incorporation reduced swelling and  $\text{KNO}_3$  loading capacity, it improved nutrient release and leaching control. Specifically, the 504 h cumulative release ratio in static water decreased by approximately 10.81%. In soil-column experiments, the cumulative leaching rates of  $\text{NO}_3^-$ -N and  $\text{K}^+$  were reduced by approximately 28.93% and 8.10%, respectively. The results indicate that nutrient release was controlled by the combined effects of polymer-network densification, biochar-assisted retention, and diffusion regulation. Among the tested formulations, ST80/CS20/BC20 showed the best balance between nutrient loading and release control. Overall, this biomass-based composite hydrogel provides a promising strategy for developing environmentally compatible slow-release fertilizers, although further studies on mechanical strength, biodegradation, crop response, and field-scale performance are still needed.

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